

Validity Analysis: MILU

Target context: Bangladeshi Bengali University Student — Story Generation Thesis Deployment

Overall risk: **HIGH**

Deployment Context

Use case: The focus is region-dependent generic llm adaptation.

Task: domain-dependent language and dialect understanding

LLMs evaluated: Generic frontier model vs small llms as well as region specific llms

Domain: General llm capability on region-dependent and context driven task understanding

Setting: Evaluating region-dependent llms to assess their capability on local-context tasks.

Target population: A native bengali speaker (a university student doing bsc thesis in story generation) from Dhaka, Bangladesh is using a regional llm to generate local context-driven stories. So it is important to have Bangladesh specific entity presence in the evaluation of the bengali part of the benchmark.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology 	1	Serious concern	high
Input Content 	1	Serious concern	high
Input Form 	4	Minor gaps	high
Output Ontology 	1	Serious concern	high
Output Content 	1	Serious concern	high
Output Form 	1	Serious concern	high
Average	1.5		

 = highest concern  = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

MILU is structurally misaligned with the Bangladeshi Bengali story generation deployment along five of six dimensions. Its ontology is India-first by explicit design [Q9, Q19] and excludes Bangladeshi national categories (1971 Liberation War, SSC/HSC, Bangladeshi universities, Ekushe Boi Mela, rural/riverine contexts). Empirical sampling confirms zero Bangladeshi cultural touchstones in 21 Bengali validation items [DATASET-D2-D7], 100% translation rate from English in the sampled Bengali validation split [DATASET-D8] (vs. 25% documented), and Bangladesh appearing only as a distractor option in non-Bengali splits [DATASET-D22, D23]. Ground-truth labels are validated entirely within Indian institutional frameworks [Q8, Q22, Q86] with no Bangladeshi annotators documented. Output form is closed MCQ accuracy [Q51], structurally incompatible with the deployment's open-ended story generation target. The only well-aligned dimension is input form (text in Bangla script [Q29, Q30, DATASET-D24]). High MILU Bengali scores would not predict — and could mislead — the deployment's intended evaluation construct.

ID	Evidence
Q8	"Sshubam Verma1 Mohammed Safi Ur Rahman Khan1,2 Vishwajeet Kumar3 Rudra Murthy3 Jaydeep Sen3 1Nilekani Centre at AI4Bharat 2Indian Institute of Technology, Madras 3IBM Research, India" (p.1)
Q9	"We designed MILU with an India-first perspective by collecting questions from various national, state, and regional exams." (p.2)
Q19	"MILU is designed as a culturally relevant benchmark to assess general problem-solving abilities and India-specific knowledge." (p.3)
Q22	"These portals typically tag questions manually with topic names and language details, and subject experts ensure the accuracy of the answers." (p.3)
Q29	"During the collection process, we exclude any reading-comprehension-style questions, images-based questions, and those with more than four answer options to ensure uniformity and consistency." (p.4)
Q30	"To remove incorrect language entries, we utilize a combination of INDICLID (Madhani et al., 2023) and Unicode-based filtering (Khan et al., 2024), ensuring that the questions are in the correct language." (p.4)
Q51	"For multiple choice questions, given k possible answer strings, we select the answer string (ai) with the highest conditional log probability, i.e., $\text{argmax}(\log P(a_i))$ " (p.5)
Q86	"We would like to thank EkStep Foundation and Nilekani Philanthropies for their generous grant towards building datasets, models, tools and other resources for Indian languages." (p.9)
D2	ফব্রুয়ারি মার্চ ২০২২ এ অনুষ্ঠিত রাজ্য বিধানসভা নির্বাচনে, আম আদম পার্টি (AAP) নমিনলখিতি কোন রাজ্যে সরকার গঠন করছে? ... পাঞ্জাব — Indian state assembly election (AAP in Punjab, 2022) — exclusively India-centric governance content
D3	নির্বাচন কমিশনারকে অপসারণ করা যত্নে পারে ... প্রধান নির্বাচন কমিশনারের সুপারিশে রাষ্ট্রপতি — Removal of India's Election Commissioner — refers to Indian constitutional provisions
D4	'মঙ্গল ভারত' কাজটি নমিনলখিতি কোন জাতীয় নেতার দ্বারা রচিত হয়েছিল ... আচার্য বিনোবা ভাবে — "Mangal Bharat" — Indian national leader Vinoba Bhave; India-specific cultural/literary content
D5	৬৫তম ফিল্মফেয়ার অ্যাওয়ার্ডস, ২০২০-এ সেরা সনিমোটোগ্রাফির জন্য পুরস্কারটি কে জিতেছেন? ... জয় ওজা — 65th Filmfare Awards (Indian film industry awards) — not Bangladeshi film industry
D6	_____ দিল্লিতে কুতুব মিনারের নির্মাণ সম্পন্ন করেছিলেন ... ইলতুতমিশ — Qutub Minar, Delhi — India-specific historical monument
D7	বাহমানি সাম্রাজ্যের প্রথম রাজধানী কী ছিল? ... গুলবার্গা — Bahamani Kingdom capital — medieval Indian/Deccan history, not Bangladeshi
D8	All 21 Bengali examples show is_translated: True — Every Bengali validation example in sample is translated from English
D22	ডিসেম্বর 2019 ৫৬ ভারতীয় ফার্মাকোপোয়া ... বাঙালিদের জন্য — Indian Pharmacopoeia — Bangladesh mentioned as a distractor option, not as subject matter
D23	ডিসেম্বর 2019-এ ভারতীয় ফার্মাকোপোয়ায় ১০০টি প্রশ্নের মধ্যে ১টি প্রশ্নে বাংলাদেশকে বিভ্রান্তি হিসেবে উল্লেখ করা হয়েছে — India Pharmacopoeia question — Bangladesh listed as distractor option2
D24	নিচের প্রশ্নে, দেওয়া চারটি বিকল্পের মধ্যে থেকে সঠিক বিকল্পটি নির্বাচন করুন যা প্রদত্ত শব্দের বিপরীত অর্থ প্রকাশ করে। ... শীতল — Antonym question in Bengali — vocabulary test

Practical Guidance

What This Benchmark Measures

MILU measures multiple-choice factual recall on Indian competitive-exam content rendered in Indic scripts, including Bengali. For this deployment, MILU's strongest residual signal is on universal STEM and Bengali grammar/vocabulary items [DATASET-D17-D21, D24-D25] — i.e., baseline academic Bengali competence. It can serve as a screening filter to distinguish

models with zero Bengali competence from those with some, but cannot measure Bangladeshi cultural grounding, register fidelity, or story-generation quality.

Construct Depth

Shallow for the deployment's target construct. MILU probes one cell of a much larger evaluation space (closed-form factual recall on Indian content). The 'culturally relevant' framing in the documentation [Q1, Q19] refers to Indian culture; for Bangladeshi cultural grounding, MILU provides effectively no signal (input_ontology, input_content, output_ontology, output_content all scored 1). The output_form gap (MCQ vs. story generation) further means even strong MILU performance does not validate generative ability.

What Else You Need

Substantial supplementation is required across input_ontology, input_content, output_ontology, output_content, and output_form. Recommended supplements: (1) BLUCK [WEB-15] and BnMMLU [WEB-18] for Bangladesh-sourced cultural and academic MCQ baselines; (2) BengaliMoralBench [WEB-8] and BanglaSocialBench [WEB-16] for Bangladeshi cultural and sociopragmatic context; (3) the Bengali Dialectal Bias Benchmark [WEB-1] for register/dialect coverage; (4) a purpose-built story-generation rubric authored and validated by native Bangladeshi annotators, modeled on BanglaSocialBench's native-speaker-written methodology, covering 1971 Liberation War, post-1971 political history through the 2024 July Uprising, Bangladeshi universities and exam systems, Bangladeshi geography (Dhaka neighborhoods, Padma/Meghna/Jamuna river systems, Cox's Bazar, Chittagong Hill Tracts), Islamic cultural register, and rural/riverine narrative contexts.

ID	Evidence
Q1	"MILU spans 8 domains and 41 subjects across 11 Indic languages, reflecting both general and culturally specific knowledge." (p.1)
Q19	"MILU is designed as a culturally relevant benchmark to assess general problem-solving abilities and India-specific knowledge." (p.3)
D17	স্থির প্রবাহ ট্রান্সফরমার _____ ধরন। ... শলে — Constant current transformer type question — domain-neutral engineering/STEM
D18	নম্নিলখতি কংনগুলি অক্সিডেশন-রডিকশন বক্রিয়া? ... উপরে সবগুলি — Oxidation-reduction reactions — universal science content
D19	হৃদয়ে পসেমকোর দ্বারা চহ্নতি হয়, ... এসএ নোড — SA node as cardiac pacemaker — universal medical content
D20	একটি অ্যাসেম্বলার যা একটি মেশিনে চলে কনিতু অন্য মেশিনে জন্ম মেশিনে কোড তরৈকিরে তাকে কবিলা হয়? ... ক্রস-অ্যাসেম্বলার — Cross-assembler definition — universal computer science
D21	দুটি পাত্র আছে X এবং Y। X-এ ১০০ মলি দুধ ... m = n — Milk-water mixture ratio problem — culturally neutral quantitative reasoning
D24	নচিরে প্ৰশ্নে, দেওয়া চারটি বকিল্পরে মধ্য থকে সেই বকিল্পটিনিব্বাচন করুন যা প্রদত্ত শব্দরে বপিরীত অর্থ প্রকাশ করে। ... শীতল — Antonym question in Bengali — vocabulary test
D25	নচিরে বাক্যটির সঠিক সক্রিয় রূপ নিব্বাচন করুন। সবাই তার চমৎকার নৃত্য পরবিশেনা দ্বারা মুগ্ধ হয়েছিল। ... তার চমৎকার নৃত্য পরবিশেনা সবাইকে মুগ্ধ করেছিল। — Active voice conversion of Bengali sentence — grammar in written Bengali
WEB-1	https://arxiv.org/abs/2603.21359
WEB-8	BengaliMoralBench's native Bangladeshi speaker consensus methodology is a model for the annotation approach this deployment requires (https://arxiv.org/abs/2511.03180)
WEB-15	BLUCK uses Bangladesh-sourced MCQ ontology as a category-redesign reference (https://arxiv.org/abs/2505.21092)
WEB-16	BanglaSocialBench's native-speaker-written, context-dependent evaluation methodology is the closest precedent for a story-generation rubric (https://arxiv.org/abs/2603.15949)
WEB-18	BnMMLU sources directly from Bangladeshi NCTB textbooks — viable alternative content provenance (https://arxiv.org/abs/2505.18951)